



IS2003-1 – INTRODUCTION TO ML

(For Minor Degree)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- Which of the following is NOT an application of Machine Learning?
A) Spam filtering
B) Weather prediction
C) Sorting numbers in ascending order
D) Speech recognition
- Supervised learning requires:
A) Only unlabeled data
B) Rewards and penalties
C) Input-output labeled pairs
D) Clusters of data
- Which of the following increases the chance of overfitting?
A) More training data
B) High variance
C) Regularization
D) Simplifying the model
- The entropy of a pure dataset (all examples in one class) is:
A) 0
B) 0.5
C) 1
D) Greater than 1
- Jaccard similarity is mainly useful for:
A) Numeric vectors
B) Binary data
C) Images
D) Regression models
- In a confusion matrix, True Positive means:
A) Predicted No, Actual No
B) Predicted Yes, Actual Yes
C) Predicted Yes, Actual No
D) Predicted No, Actual Yes
- KNN classification depends heavily on:
A) Random seed
B) Value of K
C) Entropy calculation
D) Gradient descent
- The Apriori algorithm reduces search space by:
A) Using regression
B) Using pruning based on support
C) Using backpropagation
D) Using cosine similarity
- FP-Growth differs from Apriori mainly because it:
A) Requires more iterations
B) Avoids candidate generation
C) Works only with binary data
D) Uses Euclidean distance
- A well-separated cluster means:
A) Data points belong to multiple clusters
B) Each cluster is far from others
C) Clusters overlap significantly
D) Each cluster has fuzzy membership
- In K-Means, the objective function minimized is:
A) Information Gain
B) Sum of squared distances from centroids
C) Gini Index
D) Euclidean maximum distance
- A drawback of K-Means is:
A) Works only for density-based clusters
B) Requires number of clusters in advance
C) Cannot be used for numeric data
D) Does not scale to large data

13. DBSCAN can find clusters of:
 A) Only spherical shapes
 B) Arbitrary shapes
 C) Overlapping attributes
 D) Only binary values
14. Cluster cohesion measures:
 A) Distance between clusters
 B) Compactness of points in a cluster
 C) Number of clusters formed
 D) Support of itemsets
15. Maximum Likelihood Estimation (MLE) aims to:
 A) Minimize entropy
 B) Maximize probability of observed data
 C) Minimize classification error directly
 D) Select random parameters
16. The probability distribution suitable for "coin toss outcomes" is:
 A) Gaussian
 B) Bernoulli
 C) Multinomial
 D) Poisson
17. An estimator is unbiased if:
 A) Its expected value equals the true parameter
 B) Its variance is zero
 C) It minimizes error
 D) It is computationally simple
18. Bayes' estimator minimizes:
 A) Prior probability
 B) Posterior expected loss
 C) Training error
 D) Gini index
19. Kernel density estimation is used for:
 A) Dimensionality reduction
 B) Smoothing probability distributions
 C) Decision tree pruning
 D) Classification accuracy
20. Condensed Nearest Neighbor reduces:
 A) Training set size
 B) K value
 C) Bias error
 D) Distance measure complexity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Machine Learning. Explain at least four real-world applications of ML.	8	L*2	1	1
	b) Explain Euclidean and Minkowski distances with examples. How are these used in similarity measurement?	8	L2	2	2
2.	a) Differentiate between Supervised, Unsupervised, and Reinforcement Learning with suitable examples.	8	L3	1	2
	b) Construct a confusion matrix for a binary classification problem and explain its components.	8	L3	2	1
3.	a) What is the role of bias and variance in learning? Illustrate with the help of a diagram of learning curves.	8	L2	1	1
	b) Describe the working of the ID3 decision tree induction algorithm.	8	L1	2	1
Unit – II					
4.	a) Define the problem of Association Rule Mining. Write down the steps of the Apriori algorithm.	8	L1	2	1
	b) Write down the K-Means algorithm and discuss issues related to centroid initialization.	8	L2	3	1
5.	a) Compare Apriori and FP-Growth algorithms in terms of efficiency and working principle.	8	L2	2	2
	b) Explain the DBSCAN algorithm. How does it detect clusters of arbitrary shapes?	8	L2	3	1

6.	a)	Explain different types of clusters with suitable examples	8	L2	2	2
	b)	What is cluster evaluation? Explain cohesion and separation as cluster quality measures.	8	L2	3	1

Unit – III

7.	a)	What is Maximum Likelihood Estimation (MLE)? Explain with an example using the Bernoulli distribution.	8	L1	4	1
	b)	Compare histograms, kernel density estimation, and k-NN methods for nonparametric density estimation.	8	L3	5	2
8.	a)	Describe Gaussian density estimation. How is it used in parametric classification?	8	L2	4	2
	b)	Explain the working of k-NN Classifier with an example	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
